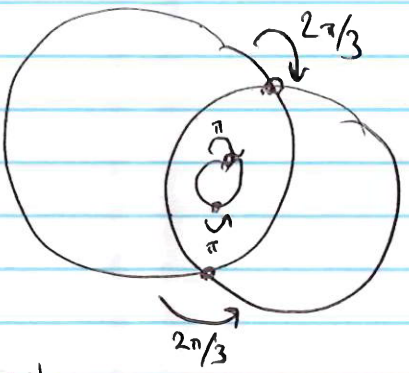
A Kleinian group is a discrete subgp of $PSL(2,\mathbb{C})$.

Γ Kleinian has 2 natural actions:

- by isometries on $\mathbb{H}^3$ s.t. $\mathbb{H}^3/\Gamma$ is a hyp. 3-orbifold.
- by conformal auts. of $S^2 = \partial\mathbb{H}^3 = \hat{\mathbb{C}}$; the set of limit pts of orbits of this action $\Lambda(\Gamma)$. Set $\Omega(\Gamma) = \hat{\mathbb{C}} \setminus \Lambda(\Gamma)$. Then $\Omega(\Gamma)/\Gamma$ is a Riemann surface (or empty).

E.g.



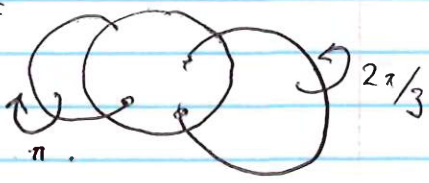
2 elliptic gens.

$\Gamma \simeq \mathbb{Z}/2\mathbb{Z} * \mathbb{Z}/3\mathbb{Z} \simeq PSL(2,\mathbb{Z})$

$\Omega(\Gamma)/\Gamma =$



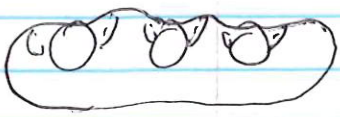
$\mathbb{H}^3/\Gamma =$



Eg.

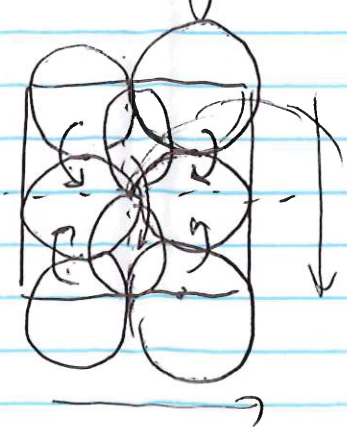


$\Gamma \simeq \mathbb{Z}^{*3}$. $\mathbb{H}^3/\Gamma =$



3 parabolic gens.

Eq.



7 parabolic gens:

$\mathbb{H}^3/\Gamma =$



If $\Omega(\Gamma)/\Gamma \neq \emptyset$, then Γ admits deformations, i.e. if $\tilde{\Gamma} \cong \Gamma$ in $PSL(2,\mathbb{C})$ then $\tilde{\Gamma}$ could be discrete & $\mathbb{H}^3/\tilde{\Gamma} \underset{\text{homeo.}}{\simeq} \mathbb{H}^3/\Gamma$.

Qn. What is the shape of the Teichmüller space in $\mathrm{Hom}(\Gamma, PSL(2,\mathbb{C}))$ containing a fixed discrete copy of Γ?

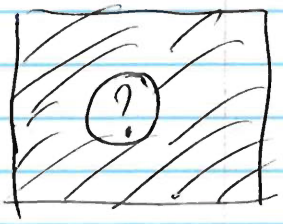
Remark. Analogous qn for $\mathbb{R}$: boring, since discrete loci are connected components of the rep^n variety.

§1. Ping-pong results.

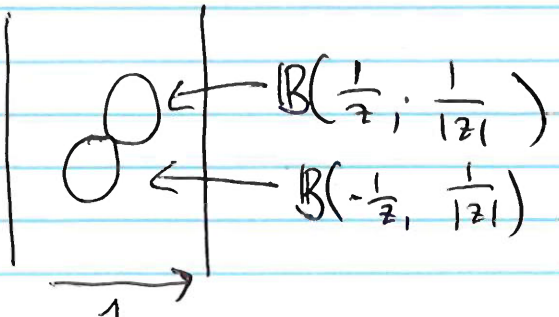
Beardon (1955): When $|z|>4$,

$$G_z = \left\langle \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ z & 1 \end{pmatrix} \right\rangle$$

is discrete & free.

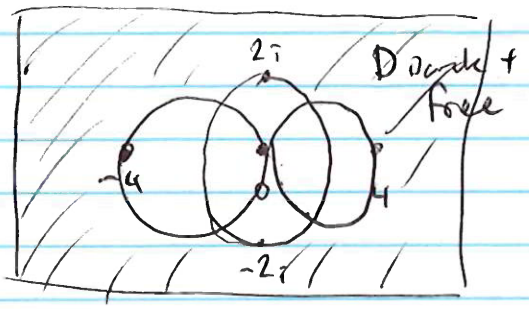


Proof: Here is a f.d. for $G_z \curvearrowright \hat{\mathbb{C}}$ when $|z| \gg 0$:



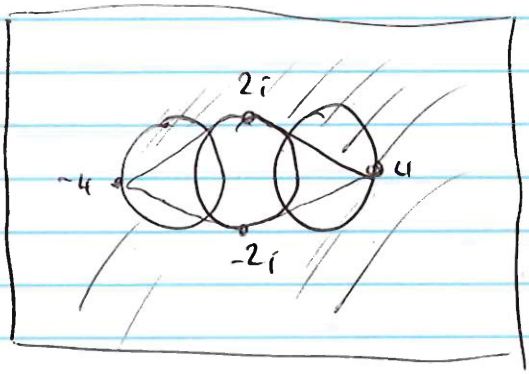
When $|z|>4$, the two disks lie inside the strip.

Chang - Jening - Ree (1958):



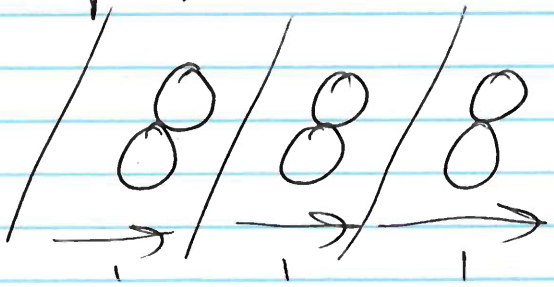
Proof:

Lyndon - Ullman (1999):



Proofs: Successively more intricate Pny-Pony arguments!

Obtain fundamental domains 'adapted' to the slope of the group, e.g.



E. - Gang - Martin - Schlenker (to appear AIF '26):
- Generalize to elliptic generators of order p, q
- Blindly following Chang-Heinz-Ree & Lyndon-Ullman gives sub-optimal results when $p \neq q$, so we also use asymmetry to further reduce the "unknown" region.

Thm. $\exists$ a (filled) dodecagon $\Omega_{p,q} \in \mathbb{C}$ s.t. if $2 \notin \Omega_{p,q}$ then

$$\left\langle \begin{pmatrix} e^{i\pi/p} & 1 \\ 0 & e^{-i\pi/p} \end{pmatrix}, \begin{pmatrix} e^{i\pi/q} & 0 \\ 2 & e^{-i\pi/q} \end{pmatrix} \right\rangle$$

is discrete & $\cong$ to $\mathbb{Z}/p\mathbb{Z} * \mathbb{Z}/q\mathbb{Z}$.

Application 1. In progress work: enumerating all arithmetic subgroups of $PSL(2,\mathbb{C})$ with two elliptic generators. If bounds on degree of trace field are known, then there are only finitely many candidate groups in any compact set. $\Omega_{p,q}$ provides such a compact set.

Application 2. A conjecture of Moree-Gerard, Oyserman, Veselov (IMRN '24):
The specialized Burau rep^n

$$B_3 \to \left\langle \begin{pmatrix} -t & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ t & -t \end{pmatrix} \right\rangle$$

is faithful outside the annulus

$$\frac{3-\sqrt{5}}{2} \leq |t| \leq \frac{3+\sqrt{5}}{2}.$$

(More generally, the specialized q-real numbers are equal to the limit set of a certain family of q-F groups).

Bands of this form are "semialgebraic neighborhoods of ∞". They do not approximate the boundary of deformation space well, & it is hard to refine them.

§2. Semialg. approximations to def. space near cusps.

Thm. (E., '25): Let G be a geom. finite Kleinian group with $\Omega(G) \neq \emptyset$. Then the qc. deformation space

$$QH(G) = \left\{ \begin{array}{l} \rho \in \mathrm{Hom}(G, PSL(2,\mathbb{C})) : \rho \text{ is discrete,} \\ \text{faithful, type-preserving \& } \rho(G) \text{ q.c.} \\ \text{conjugate to } G \end{array} \right\}$$

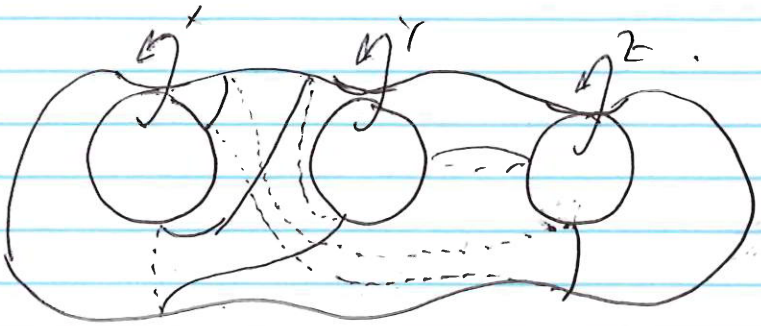
decomposes as a union of countably many computable semialgebraic sets in $\mathrm{Hom}(G, PSL(2,\mathbb{C}))$.

I sketch the construction in a special case: $\Omega(G)/G$ is a single n-punctured sphere.

STEP 1 Fix a basepoint $G_0 \in QH(G)$ such that the pleating locus of ∂ hconv $\Lambda(G)/G_0$ is a rational lamination (= cpct support).

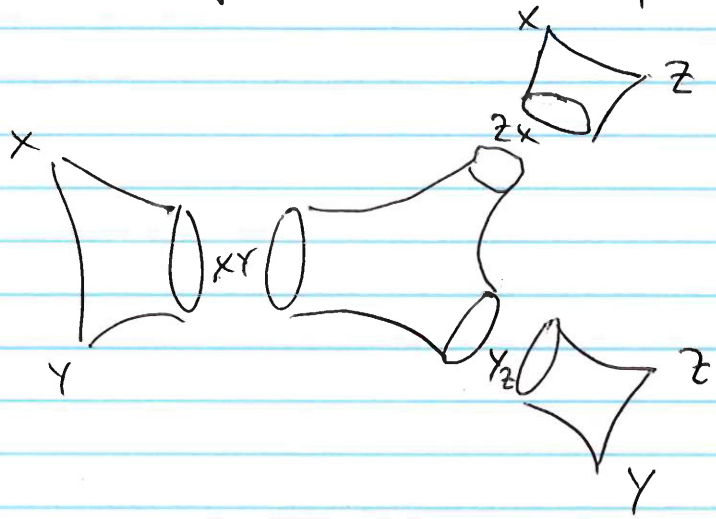
Then $\Lambda(G_0)$ cuts out a pattern of overlapping discs.

Ex.

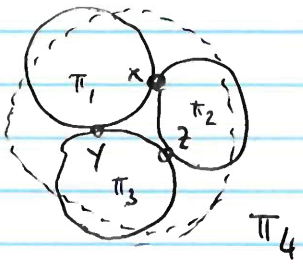


Λ is represented by XY, YZ, ZX.

The decomposition into flat pieces is:



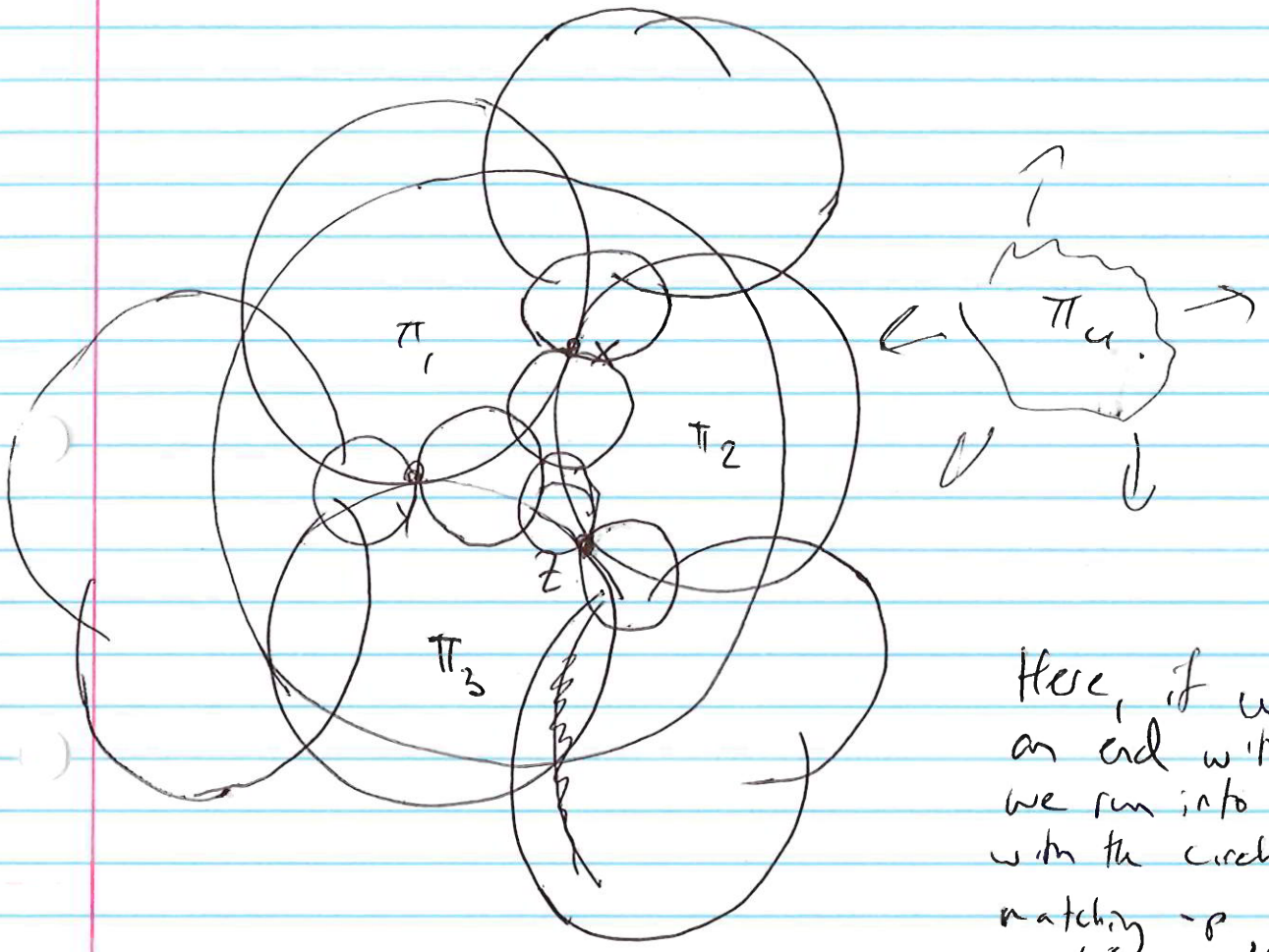
Then $\Lambda(G_0)$ cuts out



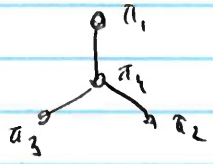
together with translates of it by scc's transverse to those in Λ.

i.e., it is a fundamental domain for the coarse structure of Γ. We refer to such a fundamental domain as a "circle chain" (c.f. Keen-Series)

STEP 2. Draw the obvious fundamental domain for each flat piece (given by isometric circles for each ∂-parallel element).



Here, if we have an end with genus, we run into trouble with the circles "not matching up around cycles". Here, we have a tree.



Step 3. Show that the domain can be made to vary algebraically (e.g. on traces) for all groups with pleating locus Λ.

Here, observe that circles in $\hat{\mathbb{C}}$ are parametrized by points in $\mathbb{P}^3_{(\mathbb{R})}$ and pencils (e.g. spanned by 2 circles) are embedded $\mathbb{P}^1$'s in $\mathbb{P}^3$.

Step 4. We have an algebraic map

$$\phi_\Lambda : \text{Hom}(G, PSL(2,\mathbb{C})) \to (\mathbb{P}^3)^N$$

which, on subsets of Hom consisting of groups with pleating locus Λ, assigns a fundamental domain to each rep^n.
putative

(We know ϕ_Λ is algebraic on this subset, and it extends uniquely.)

Write down conditions (using $\mathbb{R}$-alg. geom. of circles) in trace parameters such that the image of ϕ is a fund. dom. These conditions form a full-dimensional semialg. subset for which $\exists$ a f.d. on Ω for $\tilde{G} \in \text{Hom}(G, PSL(2,\mathbb{C}))$ with the same combinatorics.

Since the set of groups w/ rational pleating laminations is dense in $QH(G)$, we have covered $QH(G)$ with semi-alg. sets, one for each rational lamination.

Remarks - Some technicality involved to go from $(\mathbb{P}^3)^N$ to 'actual' fundamental domains, related to keeping track of twisting.

- Actually constructing non-semi-alg. sets is technically poss. but infeasible for interesting examples.

Questions for future work.

- We are essentially given f.d's for geom. fibre ends. Seems to be possible to extend (computably) to a f.d. for the whole (geom. fibre) group, folding idea of Jørgensen, Miasky.
  (Depends on technical work to appear in a different context (= tunnel #1 link gps.))
- In any case lots of flex in all these constructs. Can try to make more rigid/canonical?
- Our twisty constructs seems very related to work of ~~Blayac,~~ H

Blayac, Hamenstädt, & Marty (2025.)